SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA1403 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. *(BL3)*

Assessment Tool Weightage: 40%

Duration :60 Mins
Off Class
QUESTIONS: 5 **Total Marks:50**

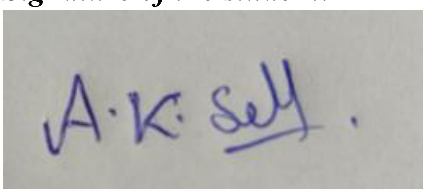
Annexure A

DESIGN TASK

Code of Conduct:

I A K Selva Prabhu (192421224) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

DESIGN TASK

S.No	Register Number	Questions
33	192421224	Design a parser to recognize Boolean expressions. **Grammar** B → B && B B → B || B B → !B B → (B) B → true B → false **Input** true && false || true **Design Requirements** (i) Implement the lexical analyzer using Lex. (ii) Implement the parser. (iii) Validate the Boolean expression. (iv) Display Accepted or Rejected. (v) Handle syntax errors.

		Design a parser to recognize variable declarations. **Grammar** D → Type IdList ; Type → int | float | char IdList → id IdList → id , IdList **Input** int a, b, c; **Design Requirements** (i) Implement the lexical analyzer using Lex. (ii) Implement the parser. (iii) Validate the declaration. (iv) Display Accepted or Rejected.

		(v) Handle syntax errors.
		Design a parser to recognize assignment statements. **Grammar** S → id = E ; E → E + T E → T T → id T → num **Input** x = a + 10; **Design Requirements** (i) Implement the lexical analyzer using Lex. (ii) Implement the parser. (iii) Validate the assignment statement. (iv) Display Accepted or Rejected. (v) Handle syntax errors.

		Design a parser to recognize function declarations.
		Grammar
		F → Type id ( ) Block
		Type → int | float | void
		Block → { }
		Input
		int display()
		{
		}
		Design Requirements
		(i) Implement the lexical analyzer using Lex.
		(ii) Implement the parser.
		(iii) Validate the function declaration.
		(iv) Display Accepted or Rejected.
		(v) Handle syntax errors.

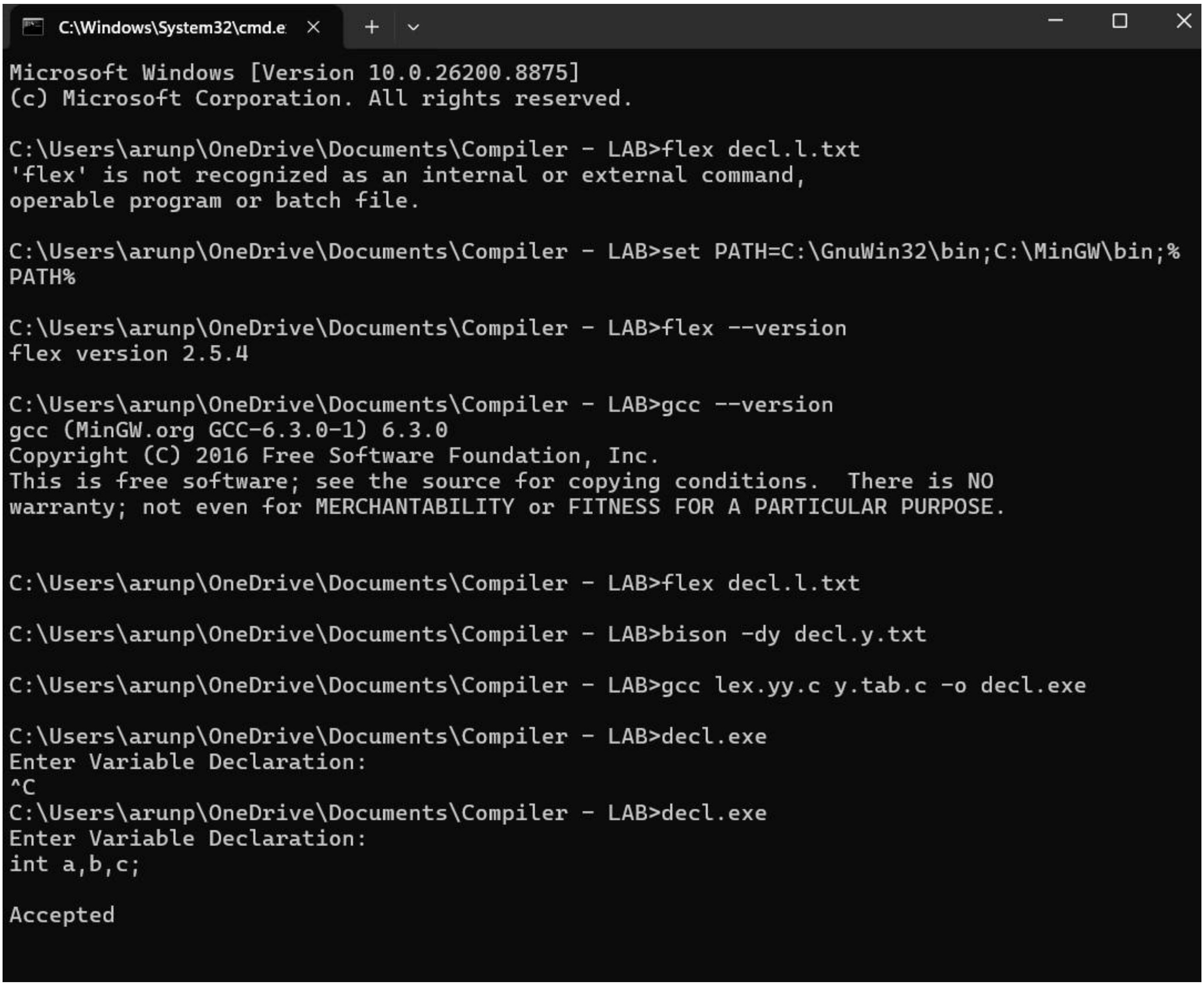
		Design a parser to recognize function calls. **Grammar** Call → id (ArgList) ArgList → id
		ArgList → id , ArgList ArgList → ε **Input** sum(a,b,c) **Design Requirements** (i) Implement the lexical analyzer using Lex. (ii) Implement the parser. (iii) Validate the function call. (iv) Display Accepted or Rejected. (v) Handle syntax errors.

1.

```
C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex --version
flex version 2.5.4

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>dir C:\GnuWin32\bin
 Volume in drive C is OS
 Volume Serial Number is BC6A-BF46

 Directory of C:\GnuWin32\bin

16-07-2026  09:07    <DIR>          .
16-07-2026  09:08    <DIR>          ..
05-05-2009  01:47           279,552 bison.exe
04-09-2022  18:36                54 bison.yacc
08-04-2004  03:26           170,496 flex++.exe
08-04-2004  03:26           170,496 flex.exe
15-03-2008  03:51         1,008,128 libiconv2.dll
07-05-2005  01:22           103,424 libintl3.dll
03-04-2009  04:45           179,200 m4.exe
24-10-2007  15:40            79,360 regex2.dll
27-12-2010  22:40            77,824 sed.exe
04-09-2022  18:36                54 yacc
              10 File(s)      2,068,588 bytes
               2 Dir(s)  326,543,646,720 bytes free

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex boolean.l
flex: can't open boolean.l

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>dir
 Volume in drive C is OS
 Volume Serial Number is BC6A-BF46

 Directory of C:\Users\arunp\OneDrive\Documents\Compiler - LAB

05-08-2026  16:41    <DIR>          .
29-07-2026  12:14    <DIR>          ..
30-07-2026  11:54            49,631 a.exe
05-08-2026  16:41               340 boolean.l.txt
05-08-2026  16:41               547 boolean.y.txt
29-07-2026  11:56               232 exp25.l.txt
29-07-2026  12:15               317 exp26.l.txt
29-07-2026  12:19               238 exp27.l.txt
29-07-2026  12:33               889 exp29.l.txt
30-07-2026  11:21            49,679 exp30.exe
30-07-2026  11:20               453 exp30.l.txt
30-07-2026  11:32               337 exp31.l.txt
30-07-2026  11:37               399 exp32.l.txt
30-07-2026  11:46               194 exp33.l.txt
30-07-2026  11:54               298 exp34.l.txt
28-07-2026  14:41               253 exp35.l.txt
28-07-2026  14:46               280 exp37.l.txt
28-07-2026  14:57               238 exp39.l.txt
28-07-2026  14:32               389 exp4.l.txt
29-07-2026  12:01            49,631 exp40.exe
28-07-2026  14:24               313 ignore.l.txt
30-07-2026  11:54            37,692 lex.yy.c
29-07-2026  12:35                79 sample.c
              21 File(s)        192,429 bytes
               2 Dir(s)  326,466,473,984 bytes free

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex boolean.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>dir lex.yy.c
 Volume in drive C is OS
 Volume Serial Number is BC6A-BF46
```

```
               2 Dir(s)  324,382,322,112 bytes free

:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\MinGW\bin;C:\GnuWin32\bin;%PATH%

:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc --version
gcc (MinGW.org GCC-6.3.0-1) 6.3.0
opyright (C) 2016 Free Software Foundation, Inc.
his is free software; see the source for copying conditions.  There is NO
arranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.


:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o boolean.exe

:\Users\arunp\OneDrive\Documents\Compiler - LAB>boolean.exe
nter Boolean Expression:
rue && false || true
^Z

ccepted

:\Users\arunp\OneDrive\Documents\Compiler - LAB>
```

2.



```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex decl.l.txt
'flex' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:\MinGW\bin;%
PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex --version
flex version 2.5.4

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc --version
gcc (MinGW.org GCC-6.3.0-1) 6.3.0
Copyright (C) 2016 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.


C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex decl.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy decl.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o decl.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>decl.exe
Enter Variable Declaration:
^C
C:\Users\arunp\OneDrive\Documents\Compiler - LAB>decl.exe
Enter Variable Declaration:
int a,b,c;

Accepted
```

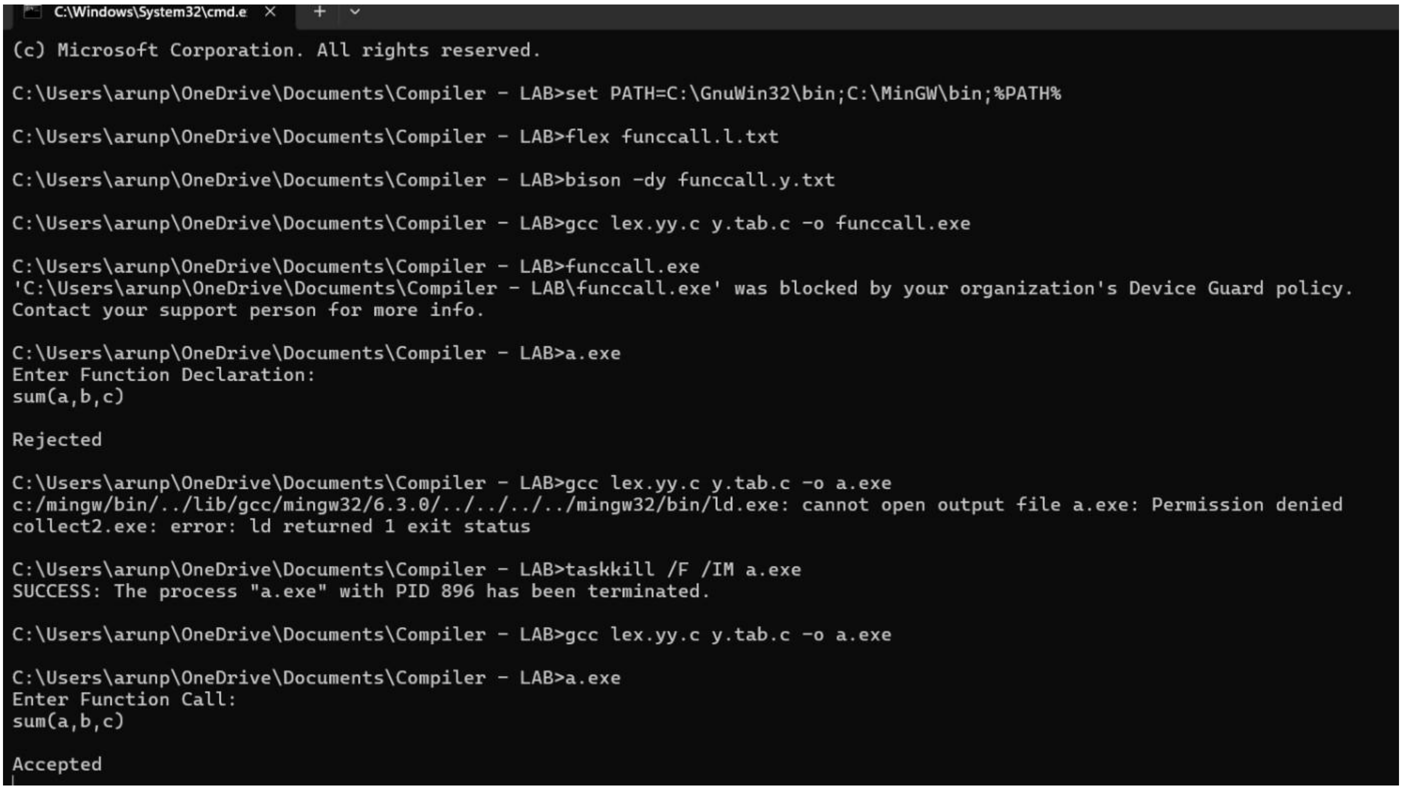
3.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:
\MinGW\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex assign.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy assign.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o ass
ign.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>assign.exe
Enter Assignment Statement:
x = a + 10;

Accepted
```

4.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:
\MinGW\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex funcdecl.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy funcdecl.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o fun
cdecl.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>funcdecl.exe
'C:\Users\arunp\OneDrive\Documents\Compiler - LAB\funcdecl.exe' was blocked
by your organization's Device Guard policy.
Contact your support person for more info.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o a.e
xe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>a.exe
Enter Function Declaration:
int display()
{
}

Accepted
```

5.



```
C:\Windows\System32\cmd.e
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:\MinGW\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex funccall.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy funccall.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o funccall.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>funccall.exe
'C:\Users\arunp\OneDrive\Documents\Compiler - LAB\funccall.exe' was blocked by your organization's Device Guard policy.
Contact your support person for more info.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>a.exe
Enter Function Declaration:
sum(a,b,c)

Rejected

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o a.exe
c:/mingw/bin/../lib/gcc/mingw32/6.3.0/../../../../mingw32/bin/ld.exe: cannot open output file a.exe: Permission denied
collect2.exe: error: ld returned 1 exit status

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>taskkill /F /IM a.exe
SUCCESS: The process "a.exe" with PID 896 has been terminated.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o a.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>a.exe
Enter Function Call:
sum(a,b,c)

Accepted
```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrat es thorough understandi ng and integrates multiple concepts effectively	Good understandin g with proper integration of most concepts	Basic understanding with partial integration	Poor understanding with no integration
Design & Implementation	30	Welldesigned solution with systematic and optimal	Correct design with minor issues	Basic design with errors or inefficiencies	Incorrect or incomplete design

		implementa tion			
Parser/Algorith m Construction	20	Accurate constructio n of parser/algo rithm with complete logic	Mostly correct construction with minor errors	Partial construction with missing logic	Incorrect or incomplete construction
Analysis & Validation	15	Insightful analysis with correct validation and justification	Good analysis with reasonable justification	Basic analysis with limited validation	Poor or incorrect analysis
Documentation & Presentation	10	Wellstructured documentat ion with diagrams, steps, and clarity	Organized documentatio n with required details	Basic documentation with missing details	Poor or incomplete documentation